Mechatronic Sensors Trainer

Fundamental Labs – IMU

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QUANSER
INNOVATE·EDUCATE

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Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

Mechatronic Sensors Trainer – Application Guide

IMU Lab

Why Explore IMUs?

Inertial Measurement Units (IMU) are key components in modern engineering systems in robotics, aerospace, autonomous systems, and more. This combination sensor can report a body's specific force and angular rate, providing valuable information about the body's acceleration as well as attitude. This also makes them useful in vibration analysis, and a key component of structural health monitoring.

Background

This lab is part of the Fundamentals labs for the Mechatronic Sensors Trainer. These labs will guide you through the sensors that are part of the board and the ones that are provided as external sensors so you can understand how to read from the sensors, what they read, when they are used, and you feel comfortable using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**006_inertial_measurement_units**

Getting Started

In this lab, you will use the **Sensors Trainer** to learn about the basics of **IMUs**. The lab will have you explore the raw data from the sensor and get a feel of the IMU's capabilities using three visualizations. Finally, you will estimate the roll & pitch of the **Sensors Trainer** using the IMU.

Before you begin this lab, ensure that the following criteria are met:

- Make sure the Sensors Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/sensors_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.